Every Algebraic Number is a Critical Value of an Integer Polynomial

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Abstract

For every irreducible $f \in \mathbb{Z}[x]$ of positive degree we construct $g \in \mathbb{Z}[x]$ and an irreducible $F \in \mathbb{Z}[x]$ with $F \mid g'$ and $F^2 \mid f \circ g$. Equivalently, every algebraic number is a critical value of an integer polynomial, so the set of critical values of $\mathbb{Z}[x]$ is exactly $\overline{\mathbb{Q}}$. The construction is explicit. The proof was assisted by AI, and has been formalized in Lean.

1 Introduction

A critical value of a nonconstant $g \in \mathbb{Z}[x]$ is a number α with $g(\beta) = \alpha$ and $g'(\beta) = 0$ for some β. Since $\overline{\mathbb{Q}}$ is closed under polynomial evaluation, every critical value is algebraic. Johannes Walcher posed the problem of establishing the reverse implication. This note proves the converse, and does so with an explicit construction.

Theorem 1. *Let f be an irreducible integral polynomial of positive degree. Then there exist F and g in $\mathbb{Z}[x]$, with F irreducible and of positive degree, such that*

$$F \mid g' \quad \textit{and} \quad F^2 \mid f \circ g.$$

The corollary is the statement in the title.

Corollary 1. *The set of critical values is exactly $\overline{\mathbb{Q}}$.*

pf. Let α be a critical value, so that $g(\beta) = \alpha$ and $g'(\beta) = 0$ for some g and β, and let $f = m_\alpha$ be the minimal polynomial of α. Then $(f \circ g)(\beta) = 0$ and $(f \circ g)'(x) = f'(g(x))\, g'(x)$, so β is a double root of $f \circ g \in \mathbb{Z}[x]$; hence $m_\beta \mid g'$ and $m_\beta^2 \mid f \circ g$. Conversely, take F and g as in Theorem 1 for $f = m_\alpha$. A root β of F has $g'(\beta) = 0$ and $f(g(\beta)) = 0$, so $g(\beta)$ is a conjugate of α; as g has integer coefficients, conjugating makes α itself a critical value. ■

2 The construction

Let f be irreducible of degree $n \geq 1$ and α a root. Write $f(x) = \sum_{k=0}^{n} a_k x^{n-k}$, so that $a_n = f(0)$. If $f = x$, then $g = x^2$ and $F = x$ work. Otherwise $f(0) \neq 0$, and since $\gcd(f, f') = 1$ in $\mathbb{Q}[x]$, the elimination ideal $(f, f') \cap \mathbb{Z}$ is nonzero; let $u, v \in \mathbb{Z}[x]$ and $\rho \in \mathbb{Z}$, $\rho \neq 0$, be such that $u\, f + v\, f' = \rho$, and set $M := \rho\, a_n$, $F(x) := f(Mx)/a_n$ and $W(x) := (v(0)F(x) - v(Mx))/\rho$. The polynomial claimed by Theorem 1 is

$$g(x) := Mx + f(Mx)\, W(x). \tag{1}$$

Theorem (restatement of Theorem 1). *F is irreducible, and $F \mid g'$ and $F^2 \mid f \circ g$.*

pf. Write $F(x) = \sum_{k=0}^{n} A_k x^{n-k}$. From $A_k = a_k M^{n-k}/a_n$ we get $A_n = 1$ and $A_k = a_k \rho^{n-k} a_n^{n-k-1}$ for $k < n$, so $F \in \mathbb{Z}[x]$ with $F(0) = 1$, which gives $W(0) = 0$. And ρ divides M and every A_k with $k < n$, so it divides $v(0)F(x) - v(Mx)$ and $W \in \mathbb{Z}[x]$. Therefore g is an integral polynomial.

Let $\beta = \alpha/M$, a root of F; therefore $g(\beta) = \alpha$, and $\rho W(\beta) = -v(\alpha)$. Then

$$\begin{aligned} g'(\alpha/M) &= M + \rho\, a_n f'(\alpha)\, W(\alpha/M) + \cancel{f(\alpha)}\, W'(\alpha/M) \\ &= M - a_n f'(\alpha) v(\alpha) = 0, \end{aligned}$$

because $u(\alpha)\,\cancel{f(\alpha)} + v(\alpha)f'(\alpha) = \rho$. Hence β is a double root of $f \circ g$, and F is primitive irreducible, so $F \mid g'$ and $F^2 \mid f \circ g$ by Gauss. ■

Any ρ will do, but $M = \rho\, a_n$ multiplies into every coefficient of F, W and g, so the reduced resultant gives the answer with the smallest coefficients.

3 Examples

Each line below is a closed identity, checkable by expansion.

Example 1 ($f = qx - p$, $p \neq 0$, $\gcd(p, q) = 1$). Here $f' = q$ and $p = qx - f$, so $(f, f') \cap \mathbb{Z} = \mathbb{Z}$ and $\rho = 1$. Choose $a, b \in \mathbb{Z}$ with $ap + bq = 1$ and take $u = -a$, $v = ax + b$; then $f(0) = -p$, $M = -p$, $F = qx + 1$, $W = x$, and

$$g = -pqx^2 - 2px, \qquad g' = -2p\,(qx+1), \qquad f \circ g = -p\,(qx+1)^2.$$

Example 2 ($f = ax^2 + bx + c$). This is irreducible of degree 2 exactly when $a \neq 0$, $\gcd(a, b, c) = 1$, and $D := b^2 - 4ac$ is not a square. Since $(f')^2 = 4af + D$,

$$\rho = \operatorname{Res}(f, f') = -aD, \qquad u = 4a^2, \qquad v = -af',$$

so $M = -acD$ and

$$F = a^3cD^2x^2 - abD\,x + 1, \qquad W = a^3bcD\,x^2 - a(b^2 - 2ac)\,x,$$

$$g = a^6bc^3D^3x^4 - 2a^4c^2(b^2 - ac)D^2x^3 + a^2bc(b^2 - ac)D\,x^2 - 2ac(b^2 - 3ac)\,x,$$

with $g' = 2ac(2a^2bcD\,x - (b^2 - 3ac))F$. For $b = 0$ the quartic term drops and $\deg g = 3 = n + 1$, the least any solution can have.

Example 3 ($f = x^3 - x - 1$). The construction gives $M = 23$ and a sextic. A hand-tuned quintic also witnesses the *statement*, and is a useful independent check of it:

$$g = -24334x^5 + 39675x^4 - 26450x^3 + 9085x^2 - 1610x + 118, \qquad F = 23x^3 - 23x^2 + 8x - 1,$$

with $g' = -230\,(23x - 7)F$ and $F^2 \mid f \circ g$, the cofactor having degree 9.

Example 4 (Dedekind's cubic). $f = x^3 - x^2 - 2x - 8$, the standard example of a field whose ring of integers is not monogenic: 2 is a common index divisor [Ded78]. With $u = 21x - 125$, $v = -7x^2 + 44x - 3$, $\rho = 1006$, $f(0) = -8$, $M = -8048$,

$$F = 65158925824x^3 + 8096288x^2 - 2012x + 1, \qquad W = -194310912x^3 + 426544x^2 + 358x,$$

$$\begin{aligned} g = {} & 1012887224144143331904x^6 - 209759614012620800x^5 \\ & - 217370176548864x^4 - 14767629312x^3 + 2350016x^2 - 10912x. \end{aligned}$$

4 Formalization

The development above has been formalized in Lean 4 [MU21] against Mathlib [The20], in about 1,200 lines. The main statement is

```
theorem main (f : Z[X]) (hf : 1 <= f.natDegree) :
    exists g H : Z[X], Irreducible H /\ H | derivative g /\ H^2 | f.comp g
```

with a strengthened form carrying $2 \leq \deg g$ and $1 \leq \deg H$, and with Corollary 1 proved from it. Every result depends only on the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.

References

[Ded78] Richard Dedekind. “Über den Zusammenhang zwischen der Theorie der Ideale und der Theorie der höheren Kongruenzen”. In: *Abhandlungen der Königlichen Gesellschaft der Wissenschaften zu Göttingen* (1878).

[The20] The mathlib Community. “The Lean Mathematical Library”. In: *Certified Programs and Proofs (CPP 2020)*. ACM, 2020.

[MU21] Leonardo de Moura and Sebastian Ullrich. “The Lean 4 Theorem Prover and Programming Language”. In: *Automated Deduction – CADE 28*. Lecture Notes in Computer Science. Springer, 2021.